Multi-Heuristic A* Campus Navigation

Optimal routes on a manually modelled VIT Vellore graph

BCSE306L Artificial Intelligence
Module 2: Problem Solving based on Searching

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Problem and AI formulation

State: road access point or junction
Action: take an outgoing path
Transition: move to its endpoint
Goal test: current node = destination
Cost: sum of positive edge distances

52 nodes and 126 directed arcs
30 landmarks and 22 junctions

Distances are illustrative model metres.
Roads and permissions need field checks.

Terminal input

Start: LIBRARY
Goal: PRP

The program prints the route, distance, expansions and time.

Type q to finish.

A* and the custom binary heap

```
frontier.push(start, priority = h(start))
while frontier has entries:
    pop the smallest f = g + h
    skip stale or worse entries
    if goal: reconstruct the path
    expand outgoing neighbors
    if candidate_g improves best_g:
        update cost and parent
        push candidate_g + h(neighbor)
```

AI decision
Choose minimum g + h, then accept a cheaper discovered route.

Heap: explicit push/pop and sift
FIFO ties by insertion number
Lazy entries and node reopening

UCS uses the same engine with h = 0. Greedy prioritises h alone.

Heuristics and shortest-path guarantees

Method	Priority	Guaranteed?
UCS / A* zero	g	Yes
Greedy Euclidean	h_E	No
A* Euclidean	g + h_E	Yes
A* scaled Manhattan	g + h_M / sqrt(2)	Yes
A* raw Manhattan	g + h_M	No

Euclidean: $h_E = scale * \sqrt{dx^2 + dy^2}$
Manhattan: $h_M = scale * (|dx| + |dy|)$

Raw Manhattan can overestimate diagonal paths. Scaling restores a valid lower bound.

Measured comparison on 50 landmark pairs

Algorithm	Optimal /50	Mean expansions	Mean gap %	Mean ms
UCS	50	30.720	0.000	0.034938
Greedy Euclidean	29	8.640	8.259	0.012857
A* Euclidean	50	13.780	0.000	0.023924
A* Scaled Manhattan	50	16.480	0.000	0.030731
A* Raw Manhattan	42	10.520	0.160	0.018959

Euclidean A*: 55.1% fewer expansions than UCS, with all 50 shortest costs.

Seed 42. Same pairs for every method. 21 repeats after one warm-up.
Mean of per-query median search times. Timings vary by machine.

Demonstration, verification and limitations

Live demonstration
Choose any IDs from --list
Find a route, then close a road
Show same-node and unreachable cases

Independent checks
13,520 campus searches
980 random directed-graph checks
Heap, reopening and heuristic audit

Limitations
Approximate access points and illustrative distances
No live GPS, crowd model or accessibility permissions
Confirm roads and calibrate scale on campus

AI declaration: OpenAI Codex drafted code, map digitisation and documents.
No external AI API performs the search. Each member must review and understand the core.